



IoT Connectivity Technologies II - Bluetooth Bluetooth

- Bluetooth is a short-range low-power communications technology for WPANs.

- Intended for replacing cables connecting portable units.
- Initially developed by Intel, Ericsson and Nokia.
- Standardized initially as IEEE 802.15.1, but afterwards Bluetooth Special Interest Group (SIG) took over the specification revisions. Latest specs are v6.0
- It is open standard, so any vendor is free to manufacture Bluetooth compatible products (but certification is required to use Bluetooth name and logo).

Technical Specs

- Bluetooth operates in the unlicensed industrial, scientific and medical (ISM) band at 2.4 to 2.4835 GHz.
- Bluetooth supports 1 Mbps data rate for version 1.2 and 3 Mbps data rate for version 2.0 combined with 'Enhanced Data Rate' mode.
- Since v4.0, there is another type of Bluetooth: **Low Energy** (or **BLE** for short) that focuses on low data rates but very low power consumption. ▪

Several aspects of the standard differ between the classic Bluetooth and BLE.

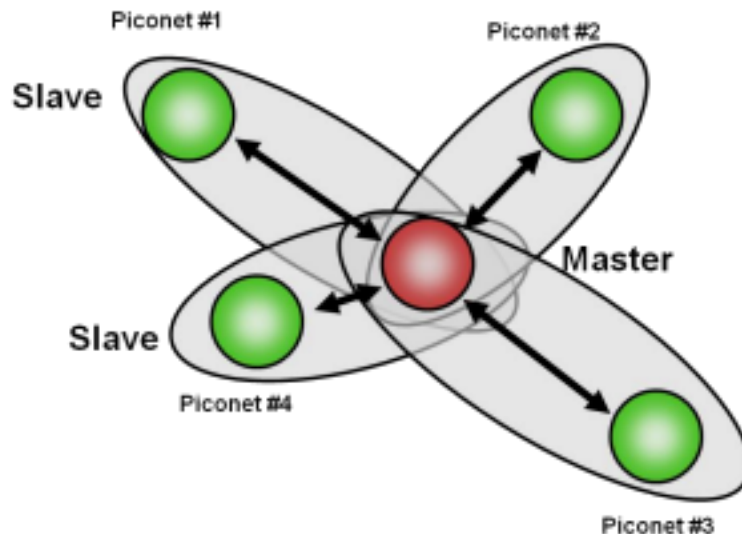
- Operating range depends on the device class:
 - Class 3 radios have a range of up to 1 meter or 3 feet
 - Class 2 radios, most commonly found in mobile devices, have a range of 10 meters or 30 feet
 - Class 1 radios, used primarily in industrial use cases, have a range of 100 meters or 300 feet

Topology - piconets

- Bluetooth devices are organized into short range networks known as **piconets**.
- The device that initializes establishment of the piconet becomes the **master**. Other devices in piconet are termed as **slaves**.
 - Provisions are in place to allow master and slave to switch their roles.
 - The simplest configuration is a point to point configuration with one master and one slave.
- **Medium Access Control:** The master is responsible for

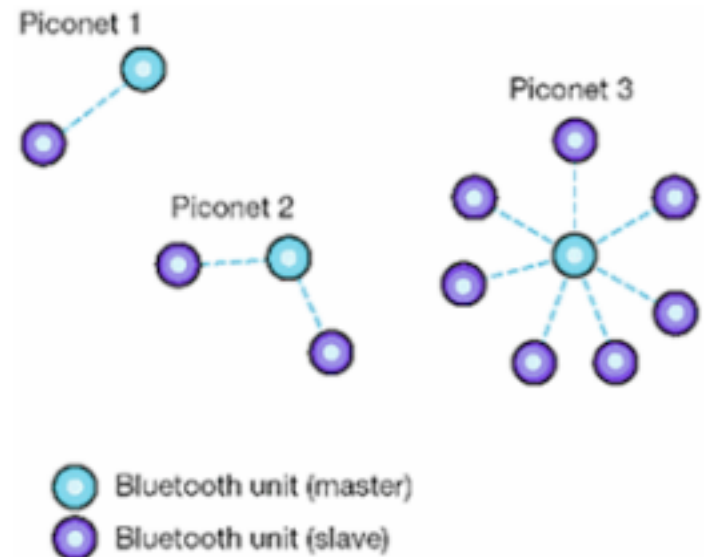
transmission control by dividing the network into a series of time slots amongst the network members, and polling each slave in turns to see if they need data transmission.

Topology - piconets



classic Bluetooth

Each device in piconet gets a 3-bit address, so there can be up to seven (non-parked) slaves and one master.



BLE mode

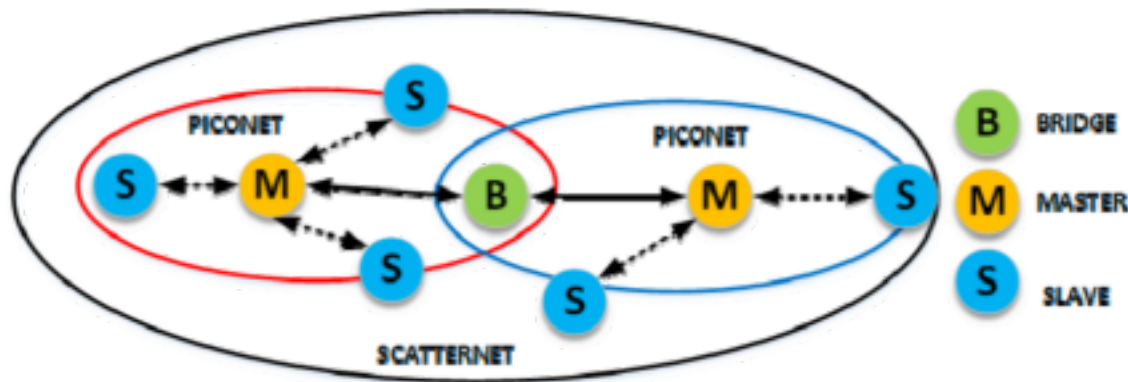
The system uses a 24-bit address and each master-slave relationship is itself a piconet.

Piconet features

- Within a piconet, the clock and unique hardware address of master determines the timing of various devices and the frequency hopping sequence of individual devices.
- Piconets are established dynamically and automatically as devices enter and leave piconets.
- There is no direct connection between the slaves. All connections are either master-to-slave or slave-to-master.
- Slaves are allowed to transmit once these have been polled by the master.
 - Transmission starts in the slave-to-master time slot immediately following a polling packet from the master.

Scatternets (classic Bluetooth)

- A device can be a member of two or more piconets. It can be a slave in one piconet and master in another. It however cannot be a master in more than one piconets.
- Devices in adjacent piconets provide a bridge to support inter-piconet connections, allowing assemblies of linked piconets to form a physically extensible communication infrastructure known as Scatternet.



Connection establishment

Inquiry run by one
Bluetooth device to
try to **discover**
other devices nearby.
Listening devices in

discoverable mode
will respond.
Process of forming
a connection
between two
Bluetooth devices.
It is called

pairing.
Once connected, a
device either actively
participates in the
network or enters a
low-power sleep
mode.

Operating modes of a connected slave

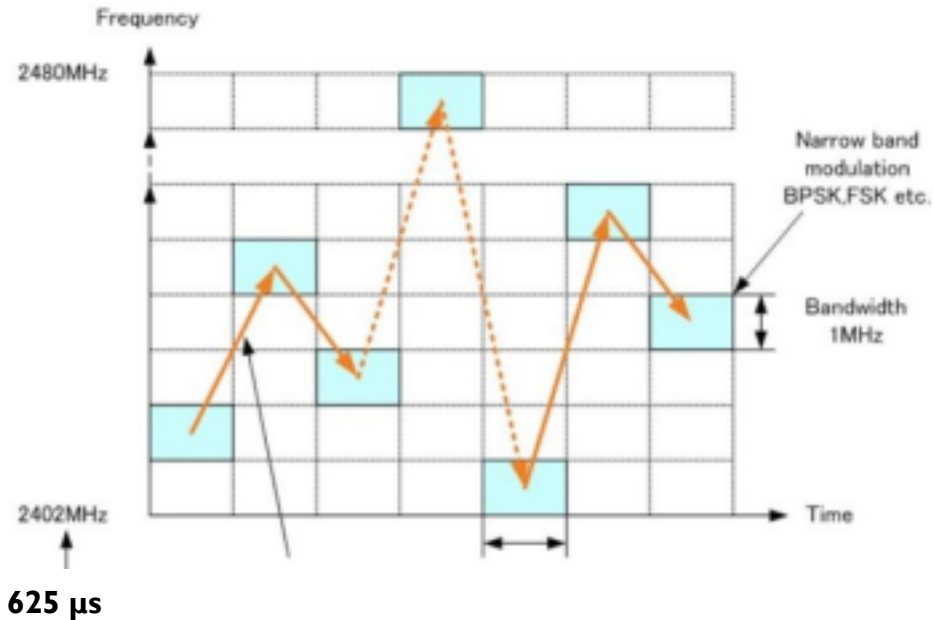
- Actively transmitting or receiving data.
- Sleeps and regularly wakes up to listens for transmissions at a set interval (e.g. 50ms).
- Power-saving mode where a master puts a device to sleep for a 'hold period'. After that period, device will return to active mode.
- Slave will become inactive until the master tells it to wake back up. • Device gives up its active-member address, and gets a parked-member address.
- Removed in Bluetooth 5.0 (because this mode is redundant after BLE).

Frequency Hopping (classic Bluetooth)

- At the physical layer, Bluetooth uses frequency hopping. Devices will continuously hop between 79 channels at a rate of

1600 hops/sec.

- A pseudorandom hop sequence is chosen by the master node. Hence each piconet has a different hop sequence.



Frequency Hopping (classic Bluetooth)

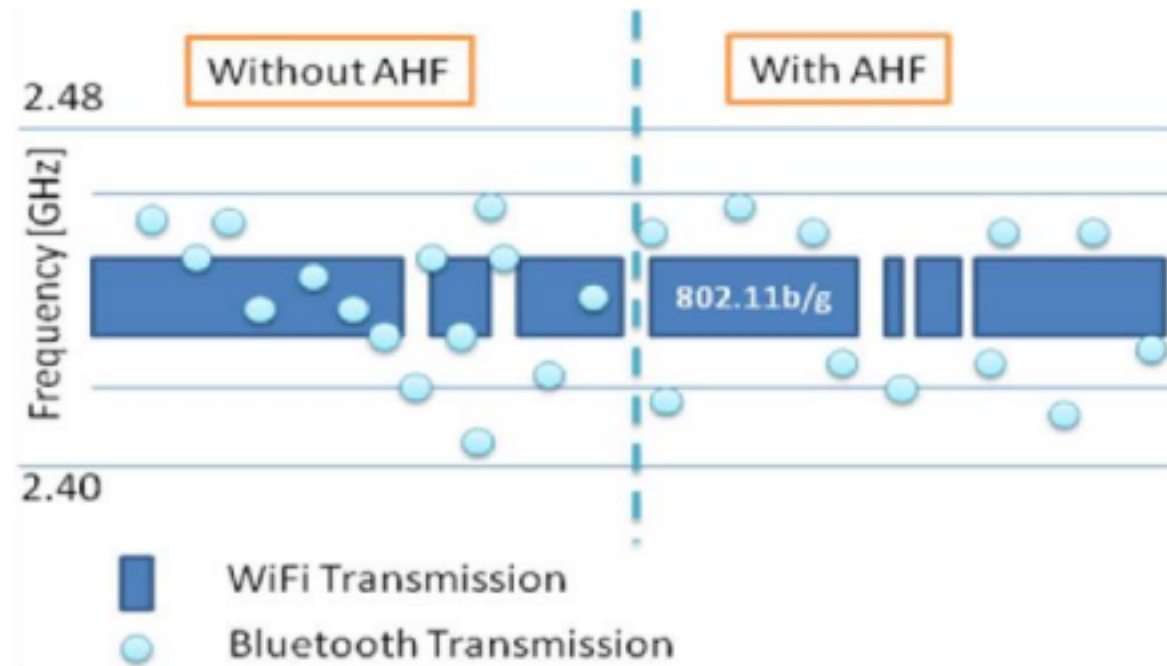
- Continuous frequency hopping allows Bluetooth devices to

get around radio interferences in the crowded 2.4GHz band.

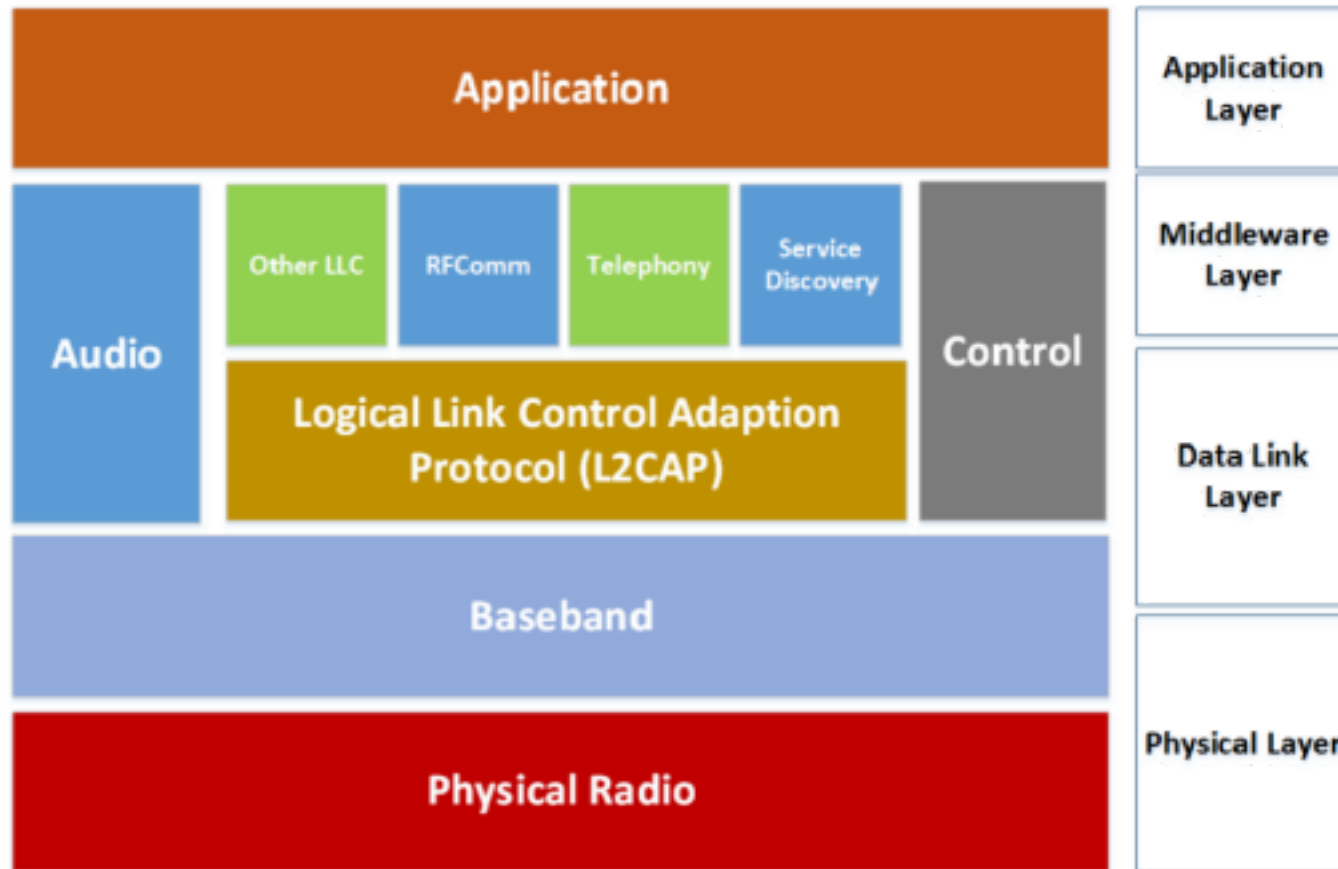
- Lots of other devices use the same band – WiFi, ZigBee, cordless phones, microwave ovens, toys, baby monitors, wireless door locks & bells, TV remote controllers, medical and fitness monitoring devices etc.
- Due to hopping, any interference will only last for one hop period ($625\mu\text{s}$), after that Bluetooth device will jump to another channel, which will likely be free of interference.

Frequency Hopping (classic Bluetooth)

- Adaptive Frequency Hopping (AFH) introduced in Bluetooth 1.2
- It allows blacklisting those channels that are facing **continuous** interference.



Protocol Stack (simplified)



Baseband layer

- Manages wireless modulation, channels (frequency hopping)

and connection procedure (inquiry, paging).

Other services include:

- Establishing logical links (connection-oriented or connection-less)
- Error detection and correction
- Data whitening – scrambling of bits within packets for (weak) security
- Challenge-response device authentication

L2CAP

- Logical Link Control and Adaptation Layer Protocol ▪

Layered over the Baseband and resides in the data link layer.

- Used to multiplex multiple logical connections between two devices.
- Provides connection-oriented and connectionless data services to upper layer protocols.
- Also provides
 - Segmentation of large packets and reassembling at receiving end
 - Monitoring QoS of traffic (delay, loss, jitter)
 - Group management – transmission to a group of devices in piconet.

RFCOMM

- It is a cable replacement protocol used for generating a virtual serial data stream.
- RFCOMM provides for binary data transport.
- Emulates UART control signals over the Bluetooth baseband layer, i.e. it is a serial port emulation.

- RFCOMM provides a simple reliable data stream to the user, similar to TCP.
- Supports up to 60 simultaneous connections between two BT devices.

Service Discovery Protocol (SDP)

- Enables applications to discover available services and their features.
- Addresses the unique characteristics of the Bluetooth environment such as, dynamic changes in the quality of services in RF proximity of devices in motion.
- Can function over a reliable packet transfer protocol.
- Uses a request/response model.

Bluetooth Applications

- Audio streaming – speakers, headphones, car stereo
- Computer accessories – keyboard, mouse
- Home automation
- File transfer between computers and/or smartphones
- Toys
- Sensor networks

BLE

- location beacons, fitness trackers, medical monitoring devices

References

- Perry Lea, Chap 5
- Misra, Chap 7
- https://www.amd.e-technik.uni-rostock.de/ma/gol/lectures/wirlec/bluetooth_info/baseband.html ▪
- <https://www.mokosmart.com/guide-on-different-bluetooth-versions/>